COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010	SECTION: CPE21S4	
DATE PERFORMED:	13/08/2026	DATE SUBMITTED:	13/08/2026
NAME:	Jullian Edrain R. Sevilla	INSTRUCTOR:	Engr. JIMLORD QUEJADO

ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.

Supplementary:

JobClass.cpp ● | PrinterClass.cpp ✕ | main.cpp ● | main.exe

PrinterClass.cpp > Printer

```cpp
#include <iostream>
#include <string>

using namespace std;

struct Job {
    int jobID;
    string userName;
    int numPages;
    Job* next;

    Job(int id, const char* name, int pages)
        : jobID(id), userName(name), numPages(pages), next(nullptr) {}
};

class Printer {
private:
    Job* front;
    Job* rear;

public:
    Printer() {
        front = rear = nullptr;
    }

    void addJob(int id, const char* name, int pages) {
        Job* newJob = new Job(id, name, pages);
        if (rear == nullptr) {
            front = rear = newJob;
        } else {
            rear->next = newJob;
            rear = newJob;
        }
        cout << "Job added: " << name << " (" << pages << " pages)\n";
    }

    void processJob() {
        if (front == nullptr) {
            cout << "No jobs to process.\n";
            return;
        }
        cout << "Processing job: " << front->userName << " (" << front->numPages << " pages)\n";
        Job* temp = front;
        front = front->next;
        if (front == nullptr) rear = nullptr;
        delete temp;
    }

```

```
48
49      void showQueue() {
50          Job* temp = front;
51          cout << "\nPending Jobs:\n";
52          while (temp != nullptr) {
53              cout << temp->jobID << " - " << temp->userName << " (" << temp->numPages << " pages)\n";
54              temp = temp->next;
55          }
56      }
57  };
58
```

JobClass.cpp ● | PrinterClass.cpp | main.cpp ● | main.exe

JobClass.cpp > Job

```
1   #include <cstring>
2
3   class Job {
4   public:
5       int jobID;
6       char userName[50];
7       int numPages;
8       Job* next;
9
10      Job(int id, const char* name, int pages) {
11          jobID = id;
12          std::strncpy(userName, name, sizeof(userName) - 1);
13          userName[sizeof(userName) - 1] = '\0';
14          numPages = pages;
15          next = nullptr;
16      }
17  };
18
```

JobClass.cpp ● PrinterClass.cpp main.cpp ● main.exe

main.cpp > main()

```cpp
#include <iostream>
#include <string>
#include <queue>

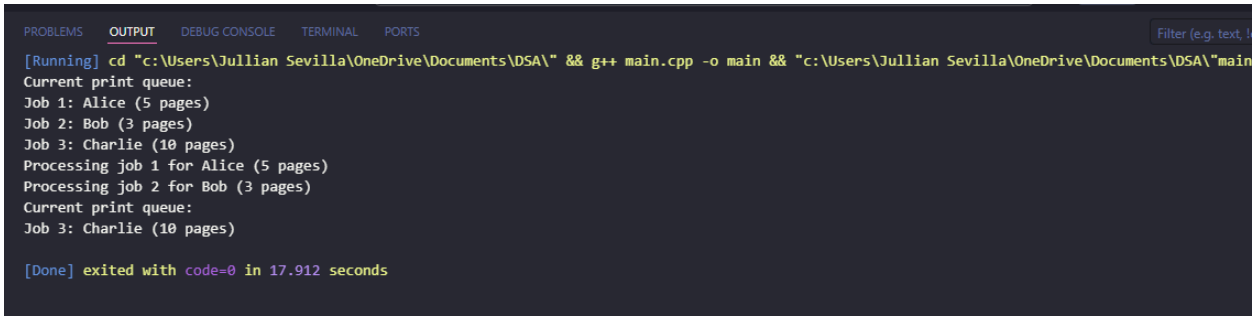

class Printer {
    struct Job { int id; std::string owner; int pages; };
    std::queue<Job> q;
public:
    void addJob(int id, const std::string &owner, int pages) {
        q.push(Job{id, owner, pages});
    }
    void showQueue() const {
        if (q.empty()) {
            std::cout << "Queue is empty\n";
            return;
        }

        std::queue<Job> copy = q;
        std::cout << "Current print queue:\n";
        while (!copy.empty()) {
            const Job &j = copy.front();
            std::cout << "Job " << j.id << ": " << j.owner << " (" << j.pages << " pages)\n";
            copy.pop();
        }
    }
    void processJob() {
        if (q.empty()) {
            std::cout << "No jobs to process\n";
            return;
        }
        Job j = q.front(); q.pop();
        std::cout << "Processing job " << j.id << " for " << j.owner << " (" << j.pages << " pages)\n";
    }
};

int main() {
    Printer printer;
    printer.addJob(1, "Alice", 5);
    printer.addJob(2, "Bob", 3);
    printer.addJob(3, "Charlie", 10);

    printer.showQueue();

    printer.processJob();
    printer.processJob();
    printer.showQueue();
    return 0;
}
```


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Filter (e.g. text,
[Running] cd "c:\Users\Jullian Sevilla\OneDrive\Documents\DSA\" && g++ main.cpp -o main && "c:\Users\Jullian Sevilla\OneDrive\Documents\DSA\"main
Current print queue:
Job 1: Alice (5 pages)
Job 2: Bob (3 pages)
Job 3: Charlie (10 pages)
Processing job 1 for Alice (5 pages)
Processing job 2 for Bob (3 pages)
Current print queue:
Job 3: Charlie (10 pages)
[Done] exited with code=0 in 17.912 seconds